

UK RAILWAY BOOKING AND SERVICE ANALYTICS

2023-2024

1. PROJECT OVERVIEW

This project focuses on analysing a railway ticketing dataset to uncover insights into customer booking behaviour, revenue trends, and train performance. The dataset contains detailed information such as ticket purchases, journey schedules, delays, cancellations, and refund requests. The goal of this analysis is to help stakeholders (railway operators, management teams, and planners) make data-driven decisions to improve operational efficiency, enhance customer experience, and maximize revenue.

2. OBJECTIVES

- ☐ Analyse total revenue and ticket sales trends.
 - ☐ Identify patterns in customer purchase behaviour (online vs station).
 - ☐ Evaluate train performance (on-time, delayed, cancelled).
 - ☐ Understand key causes of delays (e.g., signal failure, weather, staffing).
 - ☐ Analyse refund requests and their drivers.
 - ☐ Compare performance across routes, stations, and ticket types.
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3. DATA SOURCES

- ☐ **Dataset Source:** <https://mavenanalytics.io/data>
 - ☐ **Timeline:** 2023-2024
 - ☐ **Data Size:** 30,000+ rows and 18 columns
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4. DATA CLEANING (PREPROCESSING)

The following preprocessing steps were performed to ensure data quality.

- ☐ Removed duplicate records using Transaction ID.
- ☐ Corrected and changed date formats.
- ☐ Standardized categorical values (Payment Method, Ticket Type).
- ☐ Handled missing values (e.g., In delay reasons: None to “Right Away”).
- ☐ Currency symbol “£” added in the Price column.

TOOLS USED:

- ☐ Excel
 - ☐ Power BI (Power Query)
 - ☐ DAX
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5. DATA TRANSFORMATION

To enhance the quality, usability, and analytical depth of the dataset, several data transformation steps were performed using **Power Query** and **DAX**.

DEVELOPED CALCULATED TABLE:

A dedicated Calendar Table was created by the “Date of journey” column to enable time-based analysis and support DAX functions such as YTD, MTD, and trend comparisons.

The table includes:

- Year, Month Name, Quarter, Year-Quarter.

This ensured consistent date relationships and improved report performance.

DERIVED KEY PERFORMANCE INDICATORS (KPIs):

Key metrics were developed using DAX to evaluate business performance:

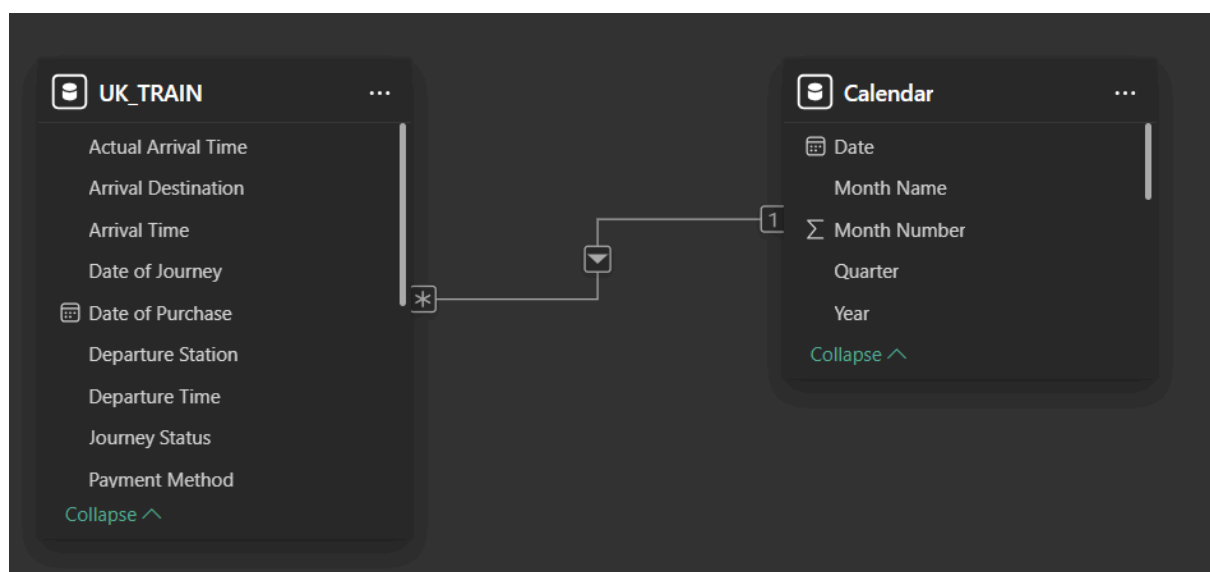
- Total Amount
- Total Bookings
- On-Time Performance (%)
- Delay Percentage (%)
- Cancellation Rate (%)
- MTD bookings

7,41,921	31653	86.82%	7.24%	5.94%	7781
Total Amount	Total Bookings	On Time %	Delay %	Cancellation %	MTD Bookings

These KPIs provided a high-level summary of operational efficiency and financial performance.

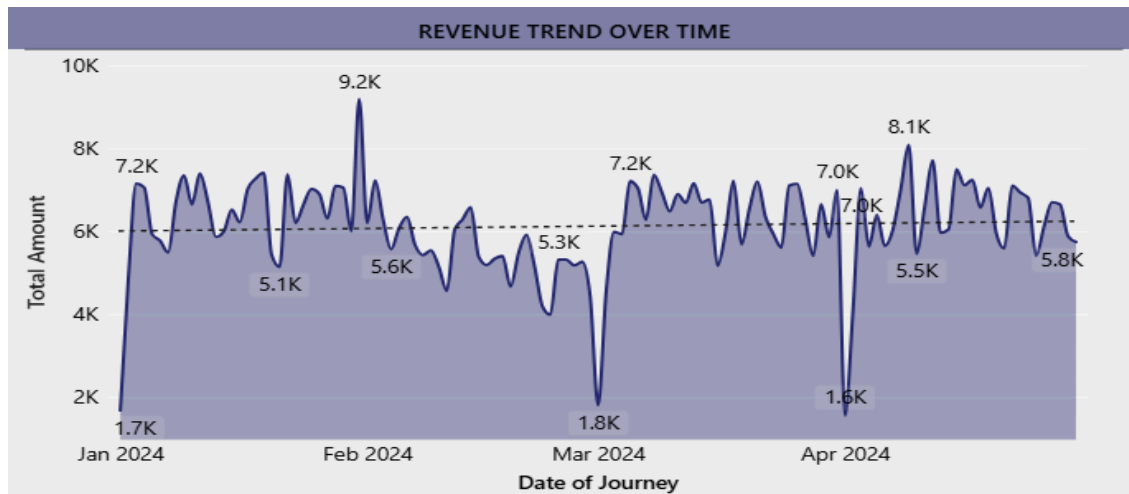
6. DATA MODELLING

Build relationships between tables to create a structured and scalable analytical data model.



7. DATA ANALYSIS (EDA) & VISUALIZATION

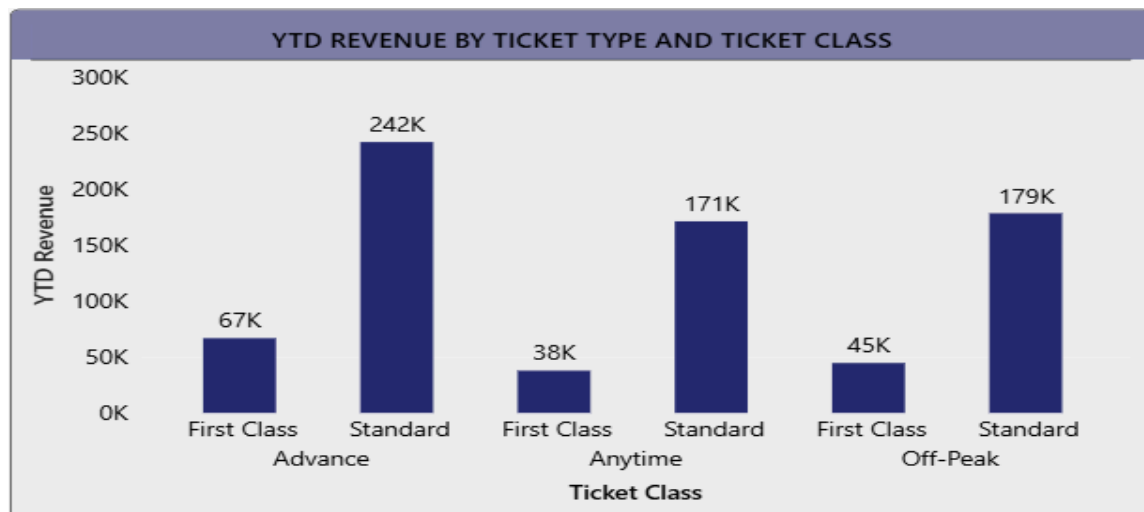
INSIGHTS BASED CHARTS:



REVENUE TREND OVER TIME – INSIGHTS:

- Descriptive Analytics: Revenue remained stable between 5K–7K with a few major peaks and sudden drops during the observed period.
- Diagnostic Analytics: Revenue fluctuations were mainly caused by high booking demand, delays, cancellations, and operational issues.
- Predictive Analytics: Revenue is likely to remain steady with occasional peaks during high-demand travel periods.
- Prescriptive Analytics: Improving on-time performance and reducing cancellations can help increase revenue consistency and customer satisfaction.

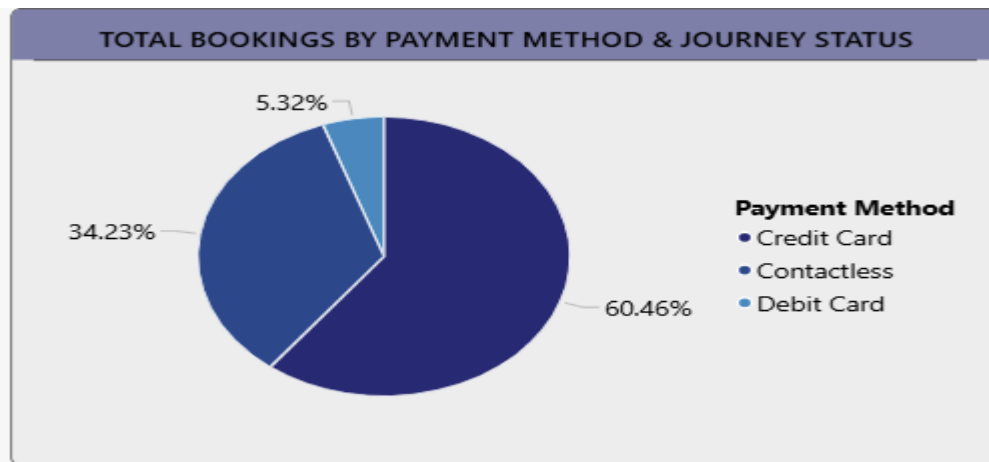
INSIGHTS BASED ON COLUMN CHART



- Descriptive Analytics: Standard Class generated the highest revenue, especially Advance tickets with 242K revenue. First Class contributed comparatively lower revenue across all ticket types.

- Diagnostic Analytics: Passengers mostly preferred affordable Standard and Advance tickets due to lower prices and better availability.
- Predictive Analytics: Standard and Advance ticket bookings are likely to continue generating the highest revenue in future periods.
- Prescriptive Analytics: Focus on promoting Advance and Off-Peak tickets and improve First Class offers to increase overall revenue.

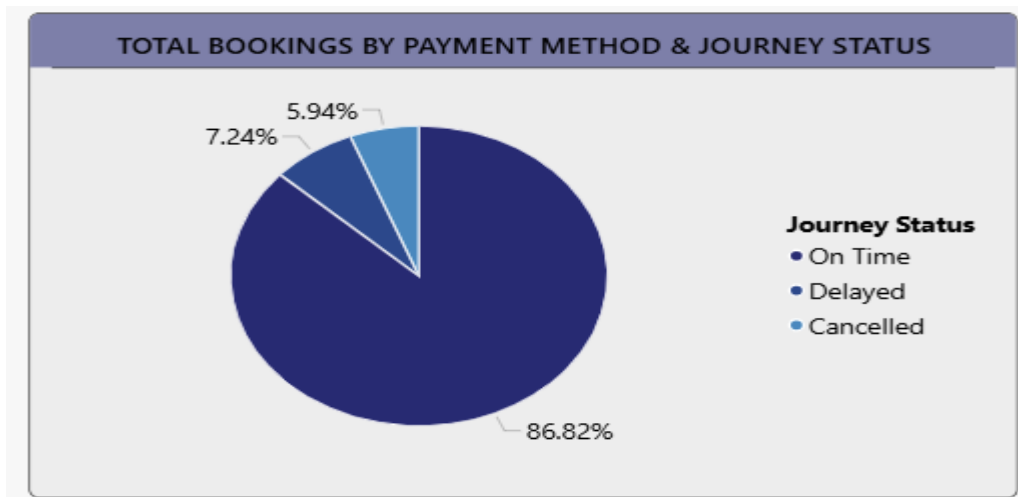
INSIGHTS BASED ON PIE CHART



PAYMENT METHOD INSIGHTS:

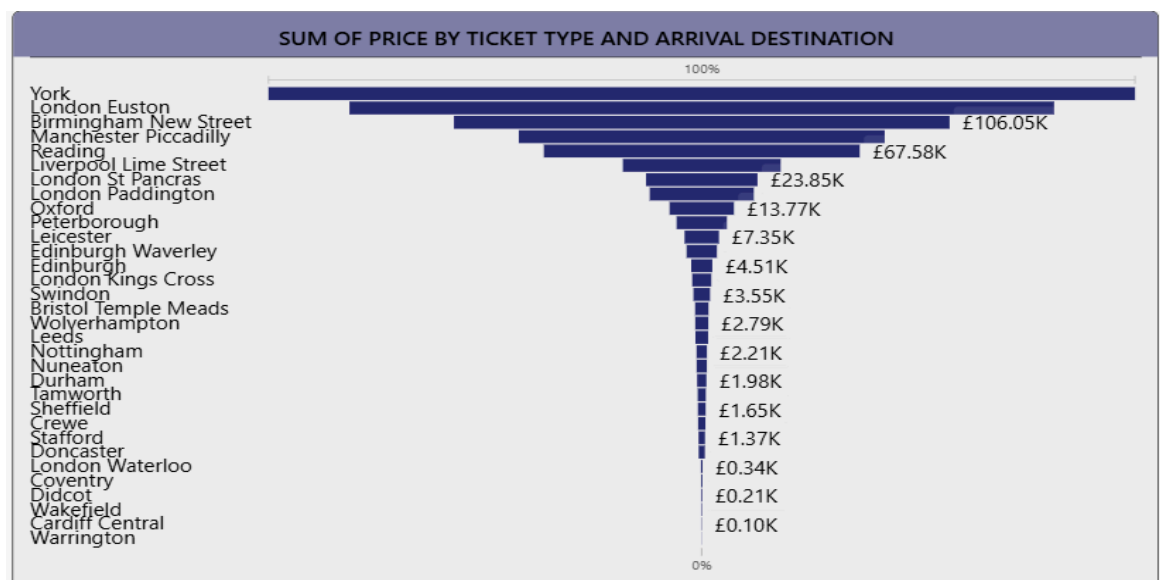
- Descriptive Analytics: Most customers used Credit Cards (60%), followed by Contactless (34%), while Debit Card usage was very low (5%).
- Diagnostic Analytics: Customers preferred Credit Cards and Contactless payments because they are faster and offer better rewards.
- Predictive Analytics: Contactless payments are likely to increase further in the future.
- Prescriptive Analytics: Provide offers and cashback to encourage more Contactless and Debit Card usage.

INSIGHTS BASED ON PIE CHART JOURNEY STATUS:



- Descriptive Analytics: Most journeys were completed On Time (86.82%), while Delayed journeys were 7.24% and Cancelled journeys were 5.94%.
- Diagnostic Analytics: High on-time performance shows efficient operations, while delays and cancellations may be caused by traffic, technical issues, or scheduling problems.
- Predictive Analytics: If current performance continues, most journeys will remain on time, but delays and cancellations may increase during peak periods.
- Prescriptive Analytics: Improve scheduling, monitor operations closely, and reduce delays/cancellations through better planning and maintenance.

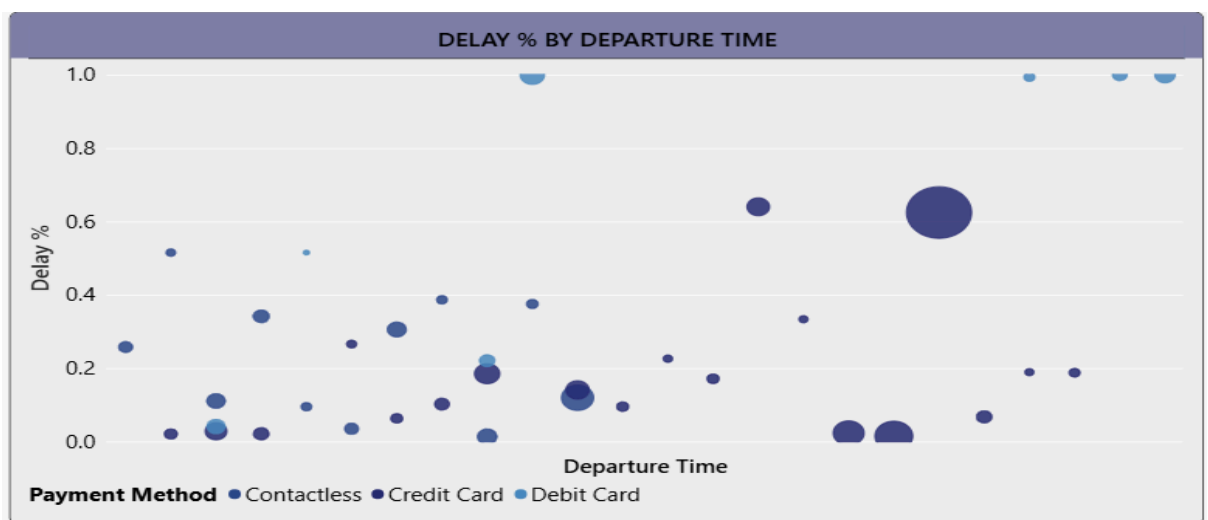
INSIGHTS BASED ON FUNNEL CHART ANALYSIS



- Descriptive Analytics: York generated the highest total revenue, followed by London Euston and Birmingham New Street. Revenue sharply decreases for other destinations, with many showing very low totals.

- Diagnostic Analytics: Major cities like York and London attract more travellers, resulting in higher ticket sales. Smaller or less frequent routes naturally bring in fewer passengers and lower revenue.
- Predictive Analytics: High-demand destinations will continue to dominate revenue in the future. Low-revenue locations are unlikely to grow significantly without targeted action.
- Prescriptive Analytics: Strengthen offerings and pricing for high-revenue routes to maximize profit. Promote lower-performing destinations through discounts, marketing, or improved services.

INSIGHTS BASED ON SCATTER PLOT ANALYSIS



- Descriptive Analytics: Delays vary across different departure times, with some times showing higher delay percentages. Larger bubbles indicate that transactions with certain payment methods (especially credit card) occur during higher delay periods.
 - Diagnostic Analytics: Certain departure times likely face more congestion, leading to increased delays. High transaction volumes at specific times may correlate with busier travel periods and operational strain.
 - Predictive Analytics: Similar departure times will likely continue showing higher delay percentages based on existing patterns. Increased passenger volume may further raise delays during peak payment/booking times.
 - Prescriptive Analytics: Adjust schedules or add buffer time for departure slots that consistently show higher delays. Improve operational efficiency during peak transaction periods to reduce delay impact.
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INSIGHTS BASED ON MAP VIEW:



- **Descriptive Analytics:** Most train journeys are concentrated around major UK cities like London, Birmingham, Manchester, and Leeds. Routes are heavily clustered in central and southern UK, with fewer trips observed in northern and far-west regions.
- **Diagnostic Analytics:** These highlighted cities are major travel hubs with higher population density and more frequent rail services. Business centres and commuter routes naturally attract more travel activity, causing dense mapping points.
- **Predictive Analytics:** Travel will continue to be concentrated around major urban centres due to sustained demand. Lesser-used routes may remain low unless new services, attractions, or developments increase travel interest.
- **Prescriptive Analytics:** Strengthen services, capacity, and connectivity between major hubs to support high demand. Promote lesser-used routes through targeted marketing, improved schedules, or better infrastructure.

FEATURES INCLUDED:

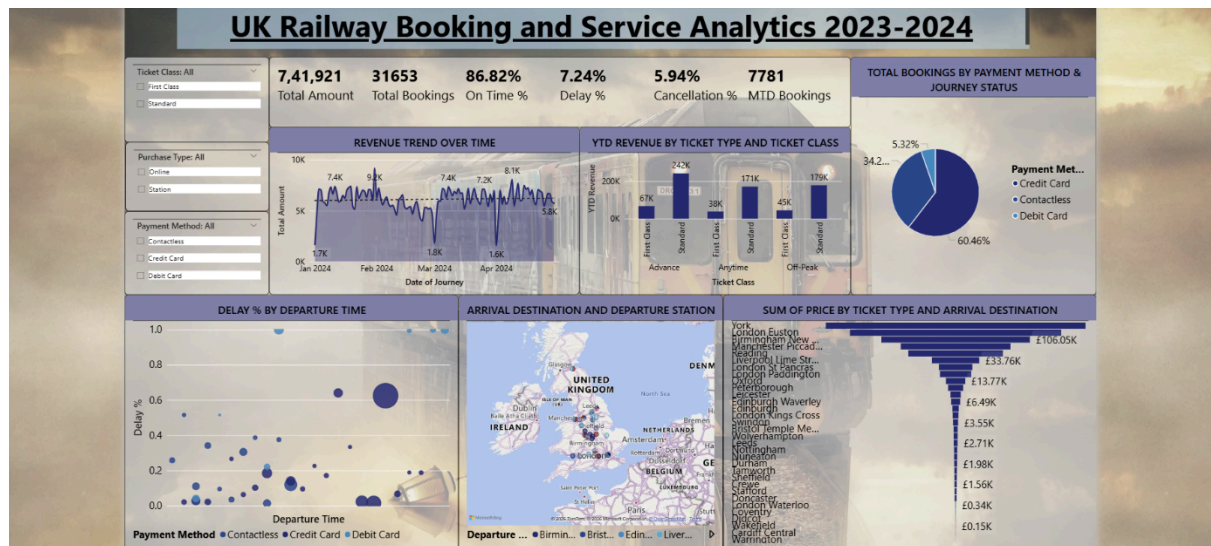
FILTERS(SLICERS):

Users can filter the data by Ticket Class, Purchase Type, and Payment Method to view specific insights and customize the dashboard based on their needs.

Drill-down Capabilities: Visuals allow users to drill down into detailed levels such as ticket type, destination, and time-based trends for deeper analysis.

Interactive Visuals: Charts, maps, and graphs respond dynamically to selections, enabling users to explore patterns, compare performance, and gain clear insights in real time.

8. FINAL DASHBOARD:



9. KEY INSIGHTS MENTIONED IN SMART WAYS

- Improve on-time performance and increase revenue by addressing delays during peak departure times and strengthening high-demand routes such as York, London Euston, and Birmingham.
- Reduce delay percentage from 7.24% to below 5%, increase on-time performance from 86.82% to 90%, and grow monthly ticket revenue by at least 10% on top-performing routes.
- Implement targeted scheduling adjustments, enhance operational coordination, and promote Off-Peak and Anytime tickets to balance demand and reduce congestion.
- These improvements align directly with the dashboard insights showing delay issues, high route concentration, and uneven revenue distribution—improving both customer satisfaction and revenue growth.
- Achieve delay reduction and service improvements within 3 months, and revenue growth targets within the next financial quarter (3–4 months).

10. CONCLUSION

The UK Railway Booking and Service dataset reveals strong booking and revenue concentration in major cities like York, London Euston, Birmingham, and Manchester. Advance tickets generate the highest revenue, indicating strong customer preference for lower-cost fare options. Overall service performance is good, with 86.82% on-time journeys, though delays still occur during specific departure times. Credit card payments dominate, reflecting a high adoption of digital payment methods. Improvements in delay management and targeted promotion of low-performing routes can enhance both customer experience and revenue.